

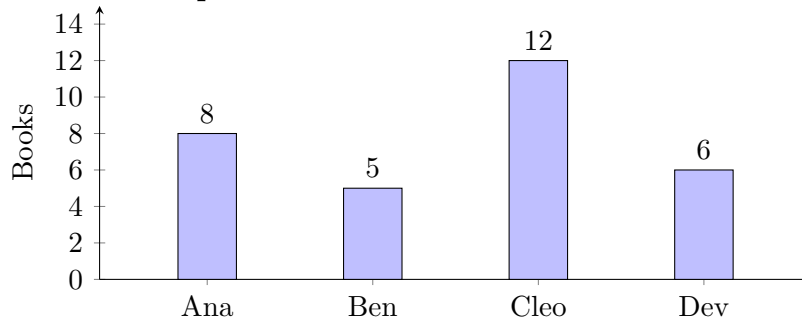
Graphs, Line Plots, and Shape Sorting

Grade 2–3 Math

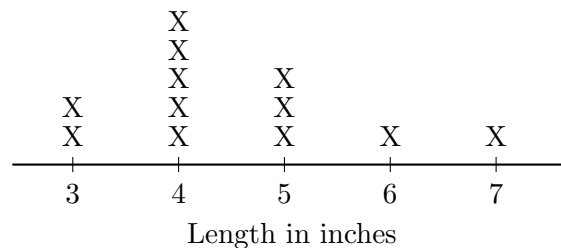
Reading a bar graph, reading a line plot, comparing and measuring data, and sorting shapes by their sides and corners. Use the three displays below.

- **Look first, then decide.** Some questions ask you to read one bar or one stack of X marks. Others ask you to add, to compare, or to sort shapes. Decide which one the question wants before you write.
- **Show how you got it.** Write the numbers you used, not just the answer.
- **Sides and corners.** A shape with 4 sides always has 4 corners. A square needs 4 equal sides *and* 4 square corners.

Graph A — Books Read in March



Line Plot B — Pencil Lengths in Ms. Ruiz’s Cup (inches)



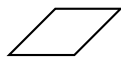
Shape Set C (no measurements shown — use the sides and corners you can see)



P



Q



R



S



T



U

1. **Graph A.** How many books did Cleo read in March?

Answer: _____

2. Graph A. How many books did all four students read altogether?

Answer: _____

3. Line Plot B. How many pencils are exactly 4 inches long?

Answer: _____

4. Line Plot B. How many pencils are in the cup altogether?

Answer: _____

5. Shape Set C. Sort the six shapes by how many sides each one has. Use the space below to keep track, then fill in the chart.

Number of sides	How many shapes
3 sides	
4 sides	
5 sides	
6 sides	

6. Line Plot B. How much longer is the longest pencil than the shortest pencil?

Answer: _____ inches

7. Graph A. How many more books did Cleo read than Ben?

Answer: _____

8. Shape Set C. Dev says, “Every shape with 4 sides is a square.” Use shapes P, Q, R and S to explain why Dev is wrong. Name one four-sided shape from the set that is *not* a square, and tell what a square needs that it does not have.

9. Line Plot B. Which pencil length is the most common one in the cup?

Answer: _____ inches

10. Line Plot B. How many pencils are longer than 4 inches? Add up every X mark to the right of the 4.

Answer: _____

11. Mia draws a square with each side 6 cm long. She also draws a rectangle that is 8 cm long and 3 cm wide.

How much greater is the distance around Mia's square than the distance around her rectangle?

Answer: _____ cm

12. Synthesis challenge — Shape Set C. Ms. Ruiz draws one more shape and adds it to Shape Set C. The new shape has 4 sides, and all 4 of its sides are the same length.

(a) There are now 7 shapes in the set. How many of them have exactly 4 sides?

(a) _____

(b) Ben says, "Now more than half of the shapes have 4 sides." Is Ben right? Use numbers to explain.